Worked solutions to

Bayesian Statistical Methods

With Applications to Machine Learning — Reich & Ghosh (2nd ed.)

Chapter 4

Linear models

Every exercise is presented in the same order:

- the **exact** exercise text from the book;
- the problem **in plain words**;
- the **background** you need, recalled from scratch;
- the **key idea** that unlocks the problem;
- a **step-by-step solution** — every algebra step written out, with a *Why?* note whenever a step is not obvious;
- a **sanity check** of the answer;
- the boxed **answer**;
- **takeaways** and common mistakes.

All numbers and figures are reproduced by `ch04/code/ch04.R` (hand-written Gibbs and Metropolis samplers, base R). The JAGS version of Exercise 9 is in `ch04/code/ch04_jags.R` and runs on GitHub. Data sources are listed in `data/README.md`.

Sai Sridhar · October 20, 2026

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Answers at a glance

1	Treatment lowers weight by ≈ 2.4 lb, $\text{Prob}(\delta < 0) \approx 0.97$; conclusion weakens under sceptical priors.
2	Bayes $\approx$ OLS under vague priors; LASSO shrinks `indus`, `age`, `rad`, `tax`; predictions for rows 501–506 run high.
3	Education is the strongest predictor; residuals heavy-tailed (Utah/Idaho); state effects cut σ from 3.9 to 2.7.
4	Each extra law: -1.4% deaths (Poisson); NB widens intervals; 0 vs 25 laws: $\approx 34{,}500$ vs $24{,}400$ — not trustworthy.
5	Female OR ≈ 12.7; 2nd/3rd class OR $0.27/0.075$; each 10 years of age OR 0.69.
6	Gamma GLM: mass grows $\approx 22\%$ per year (95% CI 7–38%).
7	$\alpha_1 \mid \cdot$ normal; $\sigma^2 \mid \cdot$, $\tau^2 \mid \cdot$ inverse-gamma (Gibbs).
8	Bed nets cut malaria odds; village effects ($\tau \approx 0.8$) widen the location-level coefficients.
9	$\rho \approx 0.99$ (near random walk); 2020 share of babies named Sophia $\approx 1.0\%$ (0.5–1.9%).
10	$g(2020) - g(1980) \approx 0.80$°C (0.69–0.91).
11	Jeffreys: big n, few p, objective; LASSO: many candidate p; informative: small new study + strong prior knowledge.
12	Sparse (spike-and-slab / horseshoe) regression; report posterior inclusion probabilities with FDR control.
13	A 3-component mixture captures the main clusters far better than one t or Laplace.
14	Logistic (occupancy) model with observer and path random effects.

A five-minute primer on the tools of this chapter

Background you need

1. Bayesian linear regression. $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\varepsilon_i \overset{\text{iid}}{\sim} \mathrm{N}(0, \sigma^2)$, with priors $\beta_j \sim \mathrm{N}(0, c^2)$ and $\sigma^2 \sim \mathrm{InvGamma}(a, b)$. Gibbs sampler:

$$\boldsymbol{\beta} \mid \sigma^2, \mathbf{Y} \sim \mathrm{N}\Big(\mathbf{V}\frac{\mathbf{X}^\top\mathbf{Y}}{\sigma^2},\ \mathbf{V}\Big), \qquad \mathbf{V} = \Big(\frac{\mathbf{X}^\top\mathbf{X}}{\sigma^2} + \frac{\mathbf{I}}{c^2}\Big)^{-1},$$

$$\sigma^2 \mid \boldsymbol{\beta}, \mathbf{Y} \sim \mathrm{InvGamma}\Big(a + \frac{n}{2},\ b + \frac{\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2}{2}\Big).$$

With c large, the posterior mean $\approx$ the least-squares estimate.

2. Standardising covariates. Using $(X_j - \bar{X}_j)/\mathrm{sd}(X_j)$ puts every coefficient on the same scale ("change in Y per standard deviation of X_j"), makes vague priors comparable across covariates and improves MCMC mixing.

3. Generalised linear models. Logistic: $\mathrm{logit}\,\mathrm{Prob}(Y_i = 1) = \mathbf{X}_i\boldsymbol{\beta}$ (e^{β_j} = odds ratio). Poisson: $\log \mathrm{E}(Y_i) = \log N_i + \mathbf{X}_i\boldsymbol{\beta}$ (e^{β_j} = rate ratio; $\log N_i$ is an *offset*). No conjugate prior exists, so we use **block random-walk Metropolis** with proposal covariance $\frac{2.4^2}{p} \times$(the classical GLM covariance) — a standard, well-tuned default.

4. Random effects. $Y_{ij} = \mathbf{X}_{ij}\boldsymbol{\beta} + \alpha_i + \varepsilon_{ij}$ with $\alpha_i \overset{\text{iid}}{\sim} \mathrm{N}(\mu, \tau^2)$: observations in the same group share α_i, so they are correlated. The model "borrows strength": groups with little data are shrunk toward μ. We let the random effects have mean μ (instead of a separate intercept) — *hierarchical centring* — which makes the sampler mix much better.

5. Posterior predictive distribution (PPD). For each posterior draw $(\boldsymbol{\beta}^{(s)}, \sigma^{(s)})$ simulate a new $Y^{*(s)} \sim f(y \mid \boldsymbol{\beta}^{(s)}, \sigma^{(s)})$. The collection $\{Y^{*(s)}\}$ is a sample from the PPD; it includes both parameter uncertainty and observation noise.

Exercise 4.1 — Does the weight-loss drug work?

Exercise 1 (exact text from the book)

A clinical trial gave six subjects a placebo and six subjects a new weight loss medication. The response variable is the change in weight (pounds) from baseline (so -2.0 means the subject lost 2 pounds). The data for the 12 subjects are:

Placebo	Treatment
2.0	-3.5
-3.1	-1.6
-1.0	-4.6
0.2	-0.9
0.3	-5.1
0.4	0.1

Conduct a Bayesian analysis to compare the means of these two groups. Would you say the treatment is effective? Is your conclusion sensitive to the prior?

In plain words

Compare the average weight change in the two groups, decide whether the drug really lowers weight, and check whether that conclusion depends on the prior.

Background you need

Two-sample normal model (Section 4.1.2): $Y_{gi} \sim \mathrm{N}(\mu_g, \sigma^2)$, $g \in \{P, T\}$. With the Jeffreys prior $\pi(\mu_P, \mu_T, \sigma^2) \propto 1/\sigma^2$,

$$\delta = \mu_T - \mu_P \mid \mathbf{Y} \;\sim\; t_{n_P+n_T-2}\Big(\bar{Y}_T - \bar{Y}_P,\; s_p^2\big(\tfrac{1}{n_P} + \tfrac{1}{n_T}\big)\Big), \qquad s_p^2 = \frac{(n_P-1)s_P^2 + (n_T-1)s_T^2}{n_P + n_T - 2}.$$

If the variances may differ, each mean has its own posterior $\mu_g \mid \mathbf{Y} \sim t_{n_g-1}(\bar{Y}_g,\; s_g^2/n_g)$, and we get δ by Monte Carlo.

Key idea

"Is the treatment effective?" becomes "what is $\mathrm{Prob}(\delta < 0 \mid \mathbf{Y})$?" (negative δ = more weight lost). Sensitivity: repeat with increasingly *sceptical* priors $\delta \sim \mathrm{N}(0, \tau^2)$ that favour "no effect".

Solution

Step 1 — summary statistics.

$\bar{Y}_P = \frac{2.0-3.1-1.0+0.2+0.3+0.4}{6} = -0.200$ and $\bar{Y}_T = \frac{-3.5-1.6-4.6-0.9-5.1+0.1}{6} = -2.600$, so $\bar{Y}_T - \bar{Y}_P = -2.400$. Pooled SD $s_p = 1.921$, standard error $s_p\sqrt{\frac{1}{6}+\frac{1}{6}} = 1.109$.

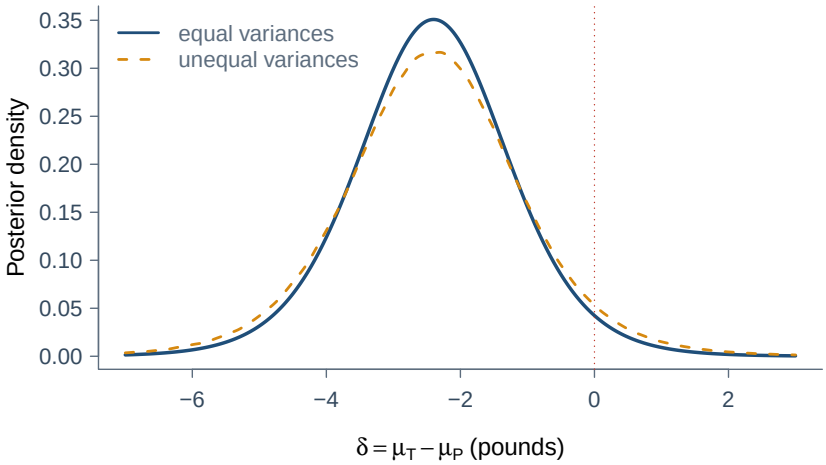
Step 2 — equal-variance posterior.

$\delta \mid \mathbf{Y} \sim t_{10}(-2.400,\; 1.109^2)$. 95% interval $-2.400 \pm t_{10,0.975} \times 1.109 = -2.400 \pm 2.228 \times 1.109 = (-4.87,\; 0.07)$, and

$$\mathrm{Prob}(\delta < 0 \mid \mathbf{Y}) = \mathrm{Prob}\Big(t_{10} < \frac{0-(-2.400)}{1.109}\Big) = \mathrm{Prob}(t_{10} < 2.164) = 0.972.$$

Step 3 — unequal variances (Monte Carlo).

Draw μ_P and μ_T from their own t posteriors (10^6 times) and subtract: mean -2.40, 95% interval $(-5.25,\ 0.45)$, $\text{Prob}(\delta < 0) = 0.957$.



› Step 4 — sensitivity to the prior on δ.

Gibbs sampler (Exercise 3.5 style) with $\delta \sim \text{N}(0, \tau^2)$:

Prior SD τ	Posterior mean of δ	95% interval	$\text{Prob}(\delta < 0 \mid \mathbf{Y})$
100 (vague)	-2.39	$(-4.88,\ 0.13)$	0.970
10	-2.38	$(-4.84,\ 0.04)$	0.973
2	-1.77	$(-3.74,\ 0.39)$	0.950
1	-0.99	$(-2.53,\ 0.68)$	0.886
0.5	-0.34	$(-1.27,\ 0.62)$	0.758

Why? A $\text{N}(0, 0.5^2)$ prior says "before the trial we were 95% sure the effect is within ± 1 pound"; six patients per group cannot overturn such a strong belief.

Sanity check

Classical two-sample t-test (one-sided): $p = 0.028$. With the vague prior, $\text{Prob}(\delta > 0 \mid \mathbf{Y}) = 0.028$ — the same number, as expected for a one-sided test with a flat prior. ✓

Answer

The treatment group lost about 2.4 lb more on average; with vague priors $\text{Prob}(\delta < 0 \mid \mathbf{Y}) \approx 0.97$, so there is fairly strong evidence the drug works, although the 95% interval just reaches 0. The conclusion **is sensitive to the prior**: under a sceptical prior concentrated near 0 (SD ≤ 1 lb) the probability drops to 0.76–0.89.

Takeaway & common mistakes

With only 12 subjects, "effective" depends on how plausible a large effect was *before* the trial. Report the posterior under several priors.

Exercise 4.2 — Boston housing: Gaussian vs. double-exponential priors

Exercise 2 (exact text from the book)

Load the classic Boston Housing Data in R:

```
> library(MASS)
> data(Boston)
> ?Boston
```

The response variable is medv, the median value of owner-occupied homes (in $1,000s), and the other 13 variables are covariates that describe the neighborhood.

(a) Fit a Bayesian linear regression model with uninformative Gaussian priors for the regression coefficients. Verify the MCMC sampler has converged, and summarize the posterior distribution of all regression coefficients.

(b) Perform a classic least squares analysis (e.g., using the lm function in R). Compare the results numerically and conceptually with the Bayesian results.

(c) Refit the Bayesian model with double exponential priors for the regression coefficients, and discuss how the results differ from the analysis with uninformative priors.

(d) Fit a Bayesian linear regression model in (a) using only the first 500 observations and compute the posterior predictive distribution for the final 6 observations. Plot the posterior predictive distribution versus the actual value for these 6 observations, and comment on whether the predictions are reasonable.

In plain words

Regress house values on 13 neighbourhood features with vague priors, compare with least squares, try a shrinkage (LASSO-type) prior, and test predictions on held-out rows.

Background you need

- Gibbs sampler for linear regression: primer, point 1.
- **Bayesian LASSO** (Park & Casella): $\beta_j \mid \sigma^2 \sim \mathrm{DE}(0, \sigma/\lambda)$, a double-exponential (Laplace) prior with a sharp peak at 0. Writing it as a normal scale mixture, $\beta_j \mid \tau_j^2 \sim \mathrm{N}(0, \sigma^2\tau_j^2)$, $\tau_j^2 \sim \mathrm{Exp}(\lambda^2/2)$, gives a Gibbs sampler: $\boldsymbol{\beta}$ normal, σ^2 inverse-gamma, $1/\tau_j^2$ inverse-Gaussian, λ^2 gamma.

Key idea

Standardise the covariates so all coefficients are on the same scale ("change in medv, in $1000s, per SD of the covariate"). Then a vague Gaussian prior reproduces least squares, while the double-exponential prior pulls weak coefficients toward 0.

Solution

(a) Model.

$\text{medv}_i = \beta_0 + \sum_{j=1}^{13} \beta_j Z_{ij} + \varepsilon_i$, Z_{ij} = standardised covariate, $\beta_j \sim \mathrm{N}(0, 100^2)$, $\sigma^2 \sim \mathrm{InvGamma}(0.01, 0.01)$. Gibbs sampler, 12,000 iterations, 2,000 burn-in.

(a) Convergence.

The draws of $\boldsymbol{\beta}$ given σ^2 are exact multivariate-normal draws, so the chain mixes almost perfectly: ESS between 9,000 and 10,000 out of 10,000 for every coefficient.

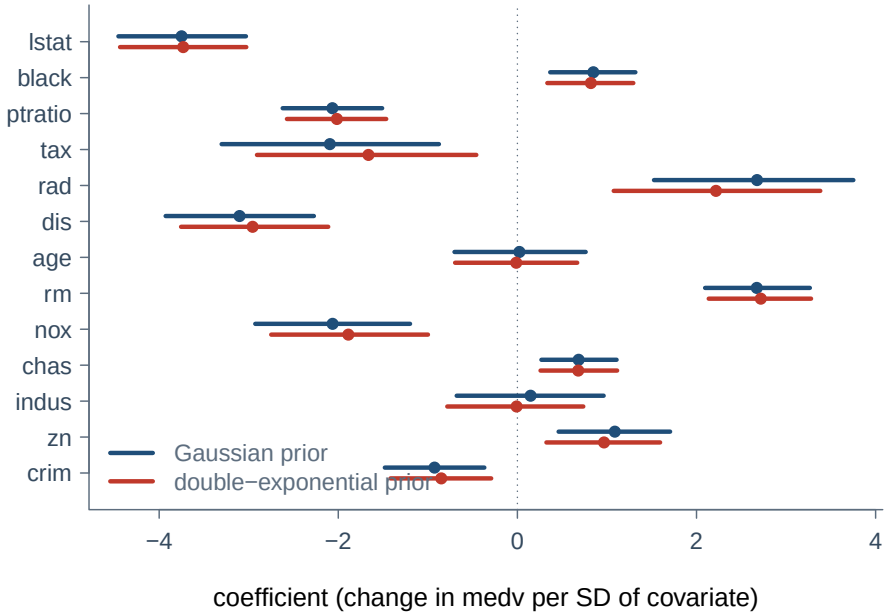
› (a)+(b) Results and comparison with least squares.

	Bayes (Gaussian prior)				Least squares	
Covariate	mean	SD	95% interval	ESS	estimate	SE
intercept	22.535	0.210	(22.13, 22.95)	10000	22.533	0.211
crim	−0.929	0.284	(−1.48, −0.37)	9909	−0.929	0.283
zn	1.086	0.318	(0.46, 1.71)	9936	1.083	0.320
indus	0.147	0.420	(−0.68, 0.96)	9003	0.141	0.422
chas	0.683	0.217	(0.27, 1.11)	9428	0.682	0.219
nox	−2.062	0.439	(−2.93, −1.20)	9726	−2.059	0.443
rm	2.676	0.298	(2.10, 3.27)	10000	2.677	0.294
age	0.023	0.376	(−0.70, 0.76)	9980	0.019	0.372
dis	−3.103	0.422	(−3.93, −2.27)	9883	−3.107	0.420
rad	2.673	0.570	(1.53, 3.75)	9987	2.665	0.578
tax	−2.090	0.626	(−3.30, −0.88)	10000	−2.079	0.634
ptratio	−2.065	0.284	(−2.62, −1.51)	9473	−2.063	0.283
black	0.848	0.244	(0.37, 1.32)	9087	0.850	0.245
lstat	−3.746	0.364	(−4.46, −3.03)	10000	−3.747	0.362

Posterior mean of σ: 4.754 (least squares: 4.745).

Numerically the two analyses agree to two or three decimals. *Conceptually* they answer different questions: least squares gives point estimates and confidence intervals (statements about repeated sampling), whereas Bayes gives a distribution for each coefficient — e.g. "given the data, there is a 95% probability that one extra SD of `lstat` lowers the median value by \$3,000–\$4,500". Strongest effects: `lstat` (% lower-status population, −), `dis` (distance to employment centres, −), `rm` (rooms, +), `rad` (+). `indus` and `age` intervals include 0.

› (c) Double-exponential (Bayesian LASSO) prior.

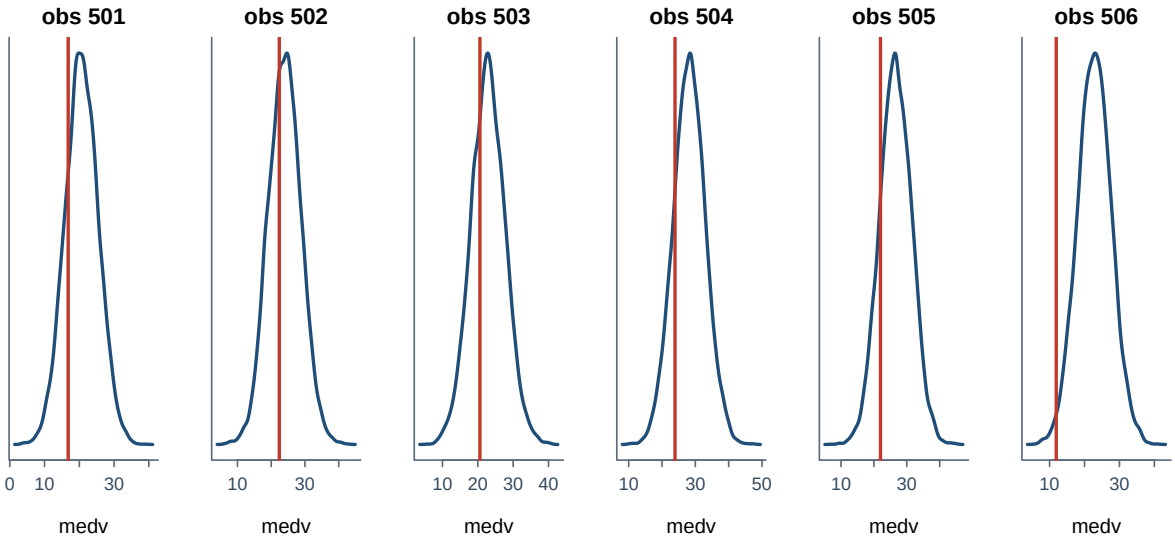


Posterior means change most for the weak or correlated covariates: `indus` $0.15 \to -0.01$, `age` $0.02 \to -0.01$ (both shrunk to ≈ 0), `rad` $2.67 \to 2.22$ and `tax` $-2.09 \to -1.67$ (these two are highly correlated, $r = 0.91$, so the prior pulls both toward 0). Strong effects (`lstat`, `rm`, `ptratio`) barely move. The posterior of λ has mean 2.97, a mild amount of shrinkage — with $n = 506$ and $p = 13$ the data dominate.

› (d) Prediction of rows 501–506.

Fit on rows 1–500; for each posterior draw compute $\mathbf{X}_i\boldsymbol{\beta}^{(s)} + \sigma^{(s)}\varepsilon^{(s)}$.

Row	Actual	PPD mean	95% prediction interval	Inside?
501	16.8	20.7	(11.3, 29.9)	yes
502	22.4	24.0	(14.8, 33.5)	yes
503	20.6	22.9	(13.5, 32.4)	yes
504	23.9	28.2	(18.8, 37.9)	yes
505	22.0	26.6	(17.2, 35.9)	yes
506	11.9	22.8	(13.5, 32.3)	**no**



Five of six actual values (red lines) fall inside their 95% intervals, which is reasonable, but *all six predictions are too high*. The last rows come from the same neighbourhood (the data are ordered by town), so they share an unmodelled local effect — the errors are not independent.

Answer

(a) The sampler mixes essentially perfectly (ESS ≈ number of draws); see table. (b) Vague-prior Bayes ≈ least squares numerically; interpretation differs (probability statements about $\boldsymbol{\beta}$). (c) The LASSO prior shrinks weak/collinear coefficients (`indus`, `age`, `rad`, `tax`) toward 0 and leaves strong ones alone. (d) 5 of 6 held-out values are inside their 95% PPD intervals, but all are over-predicted — a sign of spatial correlation.

Takeaway & common mistakes

A flat-prior Bayesian regression is "least squares with uncertainty for free"; shrinkage priors matter most when covariates are many or highly correlated.

Exercise 4.3 — What explains the 2016 swing toward the Republican candidate?

Exercise 3 (exact text from the book)

Download the 2016 Presidential Election data from the book's website. Perform Bayesian linear regression with the response variable for county i being the difference between the percentage of the vote for the Republican candidate in 2016 minus 2012 and all variables in the object X as covariates.

(a) Fit a Bayesian linear regression model with uninformative Gaussian priors for the regression coefficients, and summarize the posterior distribution of all regression coefficients.

(b) Compute the residuals $R_i = Y_i - \mathbf{X}_i\hat{\boldsymbol{\beta}}$ where $\hat{\boldsymbol{\beta}}$ is the posterior mean of the regression coefficients. Are the residuals Gaussian? Which counties have the largest and smallest residuals, and what might this say about these counties?

(c) Include a random effect for the state, that is, for a county in state $l = 1, \ldots, 50$,

$$Y_i \mid \alpha_l \sim \text{Normal}(\mathbf{X}_i\boldsymbol{\beta} + \alpha_l, \sigma^2)$$

where $\alpha_l \overset{\text{iid}}{\sim} \text{Normal}(0, \tau^2)$ and τ^2 has an uninformative prior. Why might adding random effects be necessary? How does adding random effects affect the posterior of the regression coefficients? Which states have the highest and lowest posterior mean random effect, and what might this imply about these states?

In plain words

Explain the county-level change in Republican vote share (2012 → 2016) with 10 demographic covariates, look at what the model misses, and add state effects.

Background you need

- Linear regression Gibbs sampler: primer, point 1. Random effects and hierarchical centring: primer, point 4.
- A normal distribution has skewness 0 and excess kurtosis 0; a QQ plot of normal data is a straight line.

Key idea

Counties in the same state share candidates, campaigns and local politics, so their residuals are correlated. A state random effect absorbs this shared component.

Solution

Data.

$Y_i = 100 \times$ (GOP share 2016 − GOP share 2012) in percentage points, for 3,111 counties (32 counties without valid results dropped). Covariates (standardised): population change, % aged 65+, % Black, % Hispanic, % high-school graduates, % bachelor's degree, home-ownership rate, median home value, median household income, % in poverty.

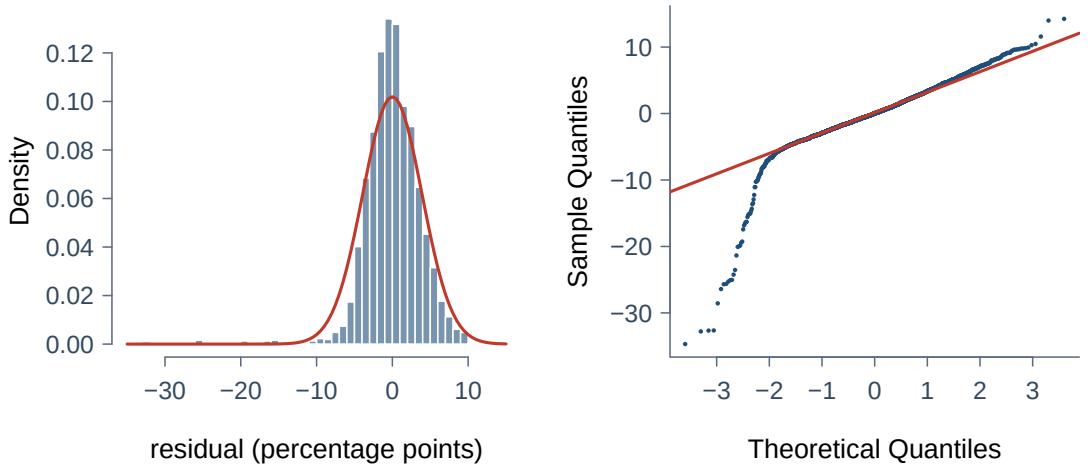
(a) Posterior summary

(Gaussian $\text{N}(0, 100^2)$ priors; coefficients per 1 SD of the covariate):

Covariate	Mean	SD	95% interval	ESS (of 5000)
intercept	3.838	0.071	(3.70, 3.98)	4992
population change 2010–14	−0.571	0.086	(−0.74, −0.40)	5000
% aged 65+	0.826	0.105	(0.62, 1.03)	4812
% Black	−1.244	0.087	(−1.42, −1.07)	5000
% Hispanic	−1.666	0.091	(−1.84, −1.49)	4964
% high-school graduates	0.041	0.133	(−0.22, 0.30)	5000
% bachelor's degree	**−3.104**	0.137	(−3.37, −2.83)	4905
home-ownership rate	−0.380	0.105	(−0.59, −0.17)	4864
median home value	−0.581	0.122	(−0.83, −0.34)	4942
median household income	1.031	0.198	(0.64, 1.42)	4711
% in poverty	0.370	0.152	(0.07, 0.67)	4874

The average county swung 3.8 points toward the Republican candidate. By far the strongest predictor is education: one SD more college graduates means a 3.1-point *smaller* swing. Counties with more Black or Hispanic residents and faster-growing populations also swung less; older counties swung more.

› (b) Are the residuals Gaussian?



No. Skewness −1.92 and excess kurtosis 13.0: a long left tail. The QQ plot bends sharply at the lower end.

Most negative residuals		Most positive residuals	
Madison County, ID	−34.7	Elliott County, KY	14.2
Utah County, UT	−32.8	McKenzie County, ND	14.0
Cache County, UT	−32.7	Clark County, MO	11.6
Davis County, UT	−32.6	Howard County, IA	10.5
Box Elder County, UT	−28.6	Starr County, TX	10.3

Interpretation. The big negative residuals are heavily Latter-day Saint (Mormon) counties in Utah and Idaho, which strongly supported Mitt Romney (a Latter-day Saint) in 2012 and gave many votes to the independent Evan McMullin in 2016 — something no demographic covariate in X captures. The positive residuals are rural, often traditionally Democratic counties (Elliott County, KY had never voted Republican before 2016) and the Hispanic border county Starr, TX, which swung unusually far toward the Republican candidate.

› (c) Why random effects?

Residuals cluster by state (state-average residuals range from −19.1 in Utah to +9.7 in DC), so counties in a state are *not* independent: home-state candidates, state-wide campaigns and third-party candidates affect

all counties in a state. Ignoring this treats 3,111 correlated counties as independent pieces of information and overstates precision.

› (c) Model.

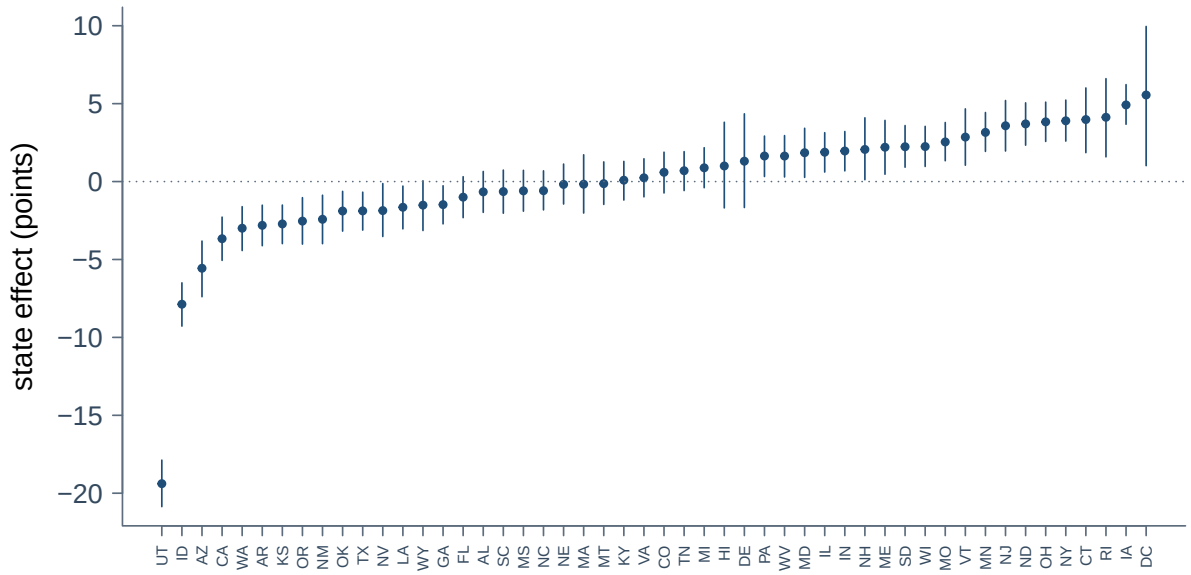
$Y_i = \mathbf{X}_i\boldsymbol{\beta} + \alpha_{s_i} + \varepsilon_i$, $\alpha_l \overset{\text{iid}}{\sim} \mathrm{N}(\mu, \tau^2)$ (with μ playing the role of the intercept), $\tau^2, \sigma^2 \sim \mathrm{InvGamma}(0.01, 0.01)$. Gibbs updates: $\boldsymbol{\beta}$ (normal), each α_l (normal, with precision $n_l/\sigma^2 + 1/\tau^2$), μ (normal), σ^2, τ^2 (inverse-gamma). ESS for the 51 state effects: 3,023–5,000 out of 5,000.

› (c) Effect on the coefficients.

	No random effects		State random effects	
Covariate	mean	SD	mean	SD
intercept / μ	3.84	0.07	3.75	0.48
population change	−0.57	0.09	−0.25	0.07
% aged 65+	0.83	0.11	0.52	0.08
% Black	−1.24	0.09	−0.85	0.08
% Hispanic	−1.67	0.09	−0.64	0.09
% high school	0.04	0.13	0.44	0.11
% bachelor's	−3.10	0.14	−3.25	0.11
home ownership	−0.38	0.11	0.07	0.08
home value	−0.58	0.12	0.26	0.11
income	1.03	0.20	−0.24	0.15
poverty	0.37	0.15	0.35	0.11

Residual SD drops from $\sigma = 3.92$ to 2.70; between-state SD $\tau \approx 4.0$. Several coefficients shrink or even change sign (Hispanic $-1.67 \to -0.64$, income $1.03 \to -0.24$, home value $-0.58 \to +0.26$): part of their apparent effect was really a *state* effect (e.g. Utah counties have high incomes and home values). The education effect is robust. The uncertainty about the overall mean grows (SD 0.07 → 0.48) because there are effectively only 51 independent states for estimating it.

› (c) State effects.



Lowest: Utah (−19.4), Idaho (−7.9), Arizona (−5.6), California (−3.7), Washington (−3.0). Highest: DC (+5.6), Iowa (+4.9), Rhode Island, Connecticut, New York (≈ +4). Interpretation: the Mountain West

(large Latter-day Saint populations) and the West Coast swung less toward the Republican candidate than their demographics predict, while Iowa and parts of the North-East swung more — consistent with the regional story of the 2016 election.

Answer

(a) Education (% bachelor's) is the dominant predictor (one SD $\Rightarrow -3.1$ points); see table. (b) Residuals are clearly non-Gaussian (heavy left tail); the worst-fitted counties are Latter-day Saint counties in Utah/Idaho (negative) and rural KY/ND/MO/IA and Starr County TX (positive). (c) State effects are needed because counties within a state are correlated; they reduce σ from 3.9 to 2.7, change several coefficients, and identify Utah/Idaho (lowest) and Iowa/North-East (highest).

Takeaway & common mistakes

Large residuals are often the most interesting part of a regression: they point to variables the model is missing (here, religion and regional politics).

Exercise 4.4 — Gun laws and homicides

Exercise 4 (exact text from the book)

Download the US gun control data from the book's website. These data are taken from the cross-sectional study in [86]. For state i, let Y_i be the number of homicides and N_i be the population.

(a) Fit the model $Y_i|\boldsymbol{\beta} \sim \text{Poisson}(N_i\lambda_i)$ where $\log(\lambda_i) = \mathbf{X}_i\boldsymbol{\beta}$. Use uninformative priors and $p = 7$ covariates in $\mathbf{X}_i$: the intercept, the five confounders Z_i, and the total number of gun laws in state i. Provide justification that the MCMC sampler has converged and sufficiently explored the posterior distribution and summarize the posterior of $\boldsymbol{\beta}$.

(b) Fit a negative binomial regression model, and compare with the results from Poisson regression.

(c) For the Poisson model in (a), compute the posterior predictive distribution for each state with the number of gun laws set to zero. Repeat this with the number of gun laws set to 25 (the maximum number). According to these calculations, how would the number of deaths nationwide be affected by these policy changes? Do you trust these projections?

In plain words

Model state death counts with a Poisson regression on five confounders and the number of gun laws; check for over-dispersion with a negative binomial; then predict national deaths under "no laws" and "all 25 laws".

Background you need

- Poisson regression with offset: $\log \mathrm{E}(Y_i) = \log N_i + \mathbf{X}_i\boldsymbol{\beta}$; e^{β_j} is a rate ratio (primer, point 3).
- **Negative binomial**: $Y \sim \mathrm{NB}(\text{mean } \mu, \text{size } r)$ has $\mathrm{Var}(Y) = \mu + \mu^2/r >$ mean. As $r \to \infty$ it becomes Poisson. It allows extra-Poisson variation (over-dispersion).

Key idea

Use block Metropolis with the GLM covariance as proposal. Check convergence with three chains ($\hat{R}$, ESS). For the projections, simulate every state's count under the new law count and add them up — one national total per posterior draw.

Solution

Data and covariates.

50 states. Confounders Z: 2009 firearm death rate, firearm-ownership quartile, unemployment quartile, non-firearm homicide quartile, firearm-export quartile. Laws: number of the 25 laws each state has (mean 5.1, SD 5.6, range 0–21). All covariates standardised; priors $\beta_j \sim \mathrm{N}(0, 10^2)$.

(a) Convergence.

Three chains $\times$ 20,000 kept iterations, acceptance 27–28% (close to the optimal 23–30% for 7 parameters); every $\hat{R} \leq 1.002$ and ESS $\approx$ 2,500 per coefficient — converged and well explored.

Coefficient (per SD)	Mean	SD	95% interval	$\hat{R}$
intercept	-9.165	0.007	$(-9.179, -9.151)$	1.000
2009 rate	0.324	0.015	(0.295, 0.352)	1.001
ownership	-0.047	0.011	$(-0.068, -0.026)$	1.001
unemployment	0.018	0.007	(0.004, 0.032)	1.002
non-firearm homicide	-0.006	0.010	$(-0.026, 0.014)$	1.002
export	0.000	0.007	$(-0.014, 0.014)$	1.001
laws	$\mathbf{-0.078}$	0.008	$(-0.094, -0.062)$	1.000

Per extra law (divide by the SD 5.61): rate ratio $e^{-0.078/5.61} = 0.986$, i.e. **1.4% fewer deaths per law** (95% CI 1.1–1.7%).

› (b) Negative binomial.

The Poisson fit is over-dispersed: Pearson $\chi^2/\text{df} = 2.70$ (should be ≈ 1). Adding a size parameter r (prior $\text{Gamma}(0.1, 0.1)$, sampled as $\log r$):

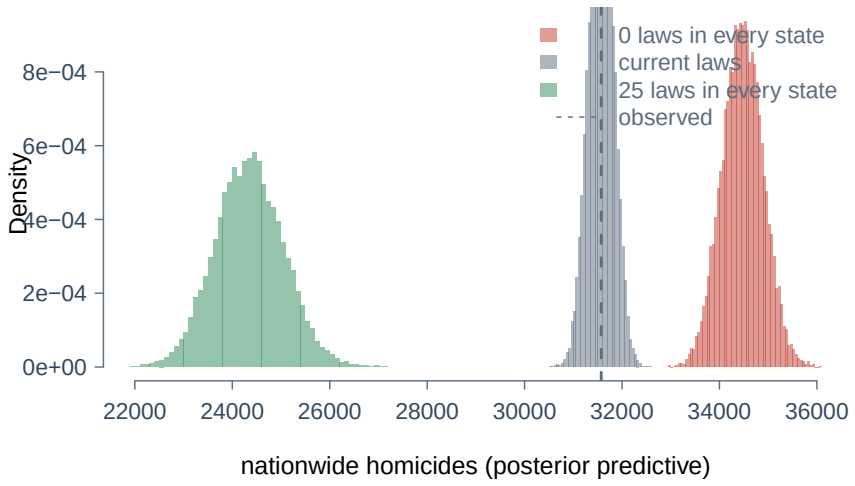
	Poisson		Negative binomial		
Coefficient	mean	SD	mean	SD	95% interval
2009 rate	0.324	0.015	0.307	0.038	(0.232, 0.382)
ownership	-0.047	0.011	-0.045	0.026	$(-0.097, 0.007)$
unemployment	0.018	0.007	0.009	0.021	$(-0.033, 0.051)$
laws	-0.078	0.008	-0.099	0.026	$(-0.151, -0.048)$

The point estimates are similar, but every SD is 2.5–3 times larger: the Poisson model was over-confident. Ownership and unemployment are no longer clearly non-zero. $r \approx 88$ (mild but real over-dispersion). The laws effect remains negative with a much wider interval.

› (c) National projections

(Poisson model, 20,000 posterior draws; each state simulated with its own confounders):

Scenario	PPD mean of national total	95% prediction interval
Current laws (check)	31,570	— (observed 31,573)
0 laws in every state	34,477	(33,656, 35,302)
25 laws in every state	24,350	(22,999, 25,741)



Taken literally: repealing all laws would add $\approx 2{,}900$ deaths a year, adopting all 25 would prevent $\approx 7{,}200$.

› (c) Do we trust these projections? No:

- **Observational, cross-sectional data**: states with many laws differ in countless unmeasured ways; the coefficient is an association, not a causal effect.
- **Extrapolation**: no state has more than 21 laws, so "25 laws everywhere" is outside the data.
- **Over-dispersion**: the Poisson intervals are far too narrow (part b).
- Counting laws assumes every law has the same effect and that effects add up on the log scale.

Sanity check

The Poisson PPD under the *current* laws reproduces the observed national total (31,570 vs. 31,573), so the simulation machinery is correct. ✓

Answer

(a) Converged ($\hat{R} \leq 1.002$, ESS $\approx 2{,}500$); each extra law is associated with 1.4% fewer deaths. (b) The NB model fits the over-dispersion ($r \approx 88$); estimates similar, SDs 2.5–3$\times$ larger. (c) $\approx 34{,}500$ deaths with 0 laws and $\approx 24{,}400$ with 25 laws vs. 31,600 now — but these projections are not trustworthy (confounding, extrapolation, over-confident Poisson intervals).

Takeaway & common mistakes

A beautifully converged MCMC sampler says nothing about whether the model answers a causal question.

Exercise 4.5 — Who survived the Titanic?

Exercise 5 (exact text from the book)

Download the titanic dataset from R,

```
library("titanic")
dat ← titanic_train
?titanic_train
```

Let $Y_i = 1$ if passenger i survived and $Y_i = 0$ otherwise. Perform a Bayesian logistic regression of the survival probability onto the passenger's age, gender (dummy variable) and class (two dummy variables). Summarize the effect of each covariate.

In plain words

Model the probability of surviving the Titanic as a function of age, sex and ticket class.

Background you need

Logistic regression: $\log \frac{p_i}{1-p_i} = \mathbf{X}_i\boldsymbol{\beta}$. Log-likelihood $\sum_i \{Y_i\eta_i - \log(1+e^{\eta_i})\}$ with $\eta_i = \mathbf{X}_i\boldsymbol{\beta}$. e^{β_j} is the **odds ratio** for a one-unit change in X_j; probabilities come from $p = \frac{e^\eta}{1+e^\eta}$.

Key idea

Dummy coding: `female` = 1 for women; `class2`, `class3` = 1 for 2nd/3rd class (1st class is the baseline). Age is centred at 30 and measured in decades.

Solution

Data.

891 passengers; 177 have missing age, leaving **714 complete cases** (survival rate 40.6%). (Dropping incomplete cases assumes age is missing at random; imputing age inside the MCMC would be the fuller Bayesian approach.)

Model.

$\text{logit}\, p_i = \beta_0 + \beta_1 \frac{\text{age}_i - 30}{10} + \beta_2\, \text{female}_i + \beta_3\, \text{class2}_i + \beta_4\, \text{class3}_i$, $\beta_j \sim \text{N}(0, 10^2)$. Block Metropolis, three chains $\times$ 16,000 kept draws, acceptance 28%, all $\hat{R} \le 1.002$.

	Mean	95% interval	Odds ratio	OR 95% interval	GLM
intercept	0.140	$(-0.26, 0.54)$	—	—	0.145
age (per 10 yr)	-0.373	$(-0.53, -0.23)$	0.689	(0.59, 0.80)	-0.370
female	2.541	(2.14, 2.95)	**12.7**	(8.5, 19.1)	2.523
2nd class	-1.313	$(-1.85, -0.76)$	0.269	(0.16, 0.47)	-1.310
3rd class	-2.596	$(-3.15, -2.04)$	0.075	(0.04, 0.13)	-2.581

Turning odds into probabilities

(30-year-old passenger; posterior median and 95% interval):

	Man	Woman
1st class	0.54 (0.44, 0.63)	0.94 (0.90, 0.96)
2nd class	0.24 (0.17, 0.31)	0.80 (0.72, 0.86)
3rd class	0.08 (0.06, 0.11)	0.52 (0.43, 0.61)

Why? For a 30-year-old man in 1st class, $\eta = \beta_0$, so $p = \frac{e^{0.14}}{1+e^{0.14}} = 0.535$; a woman adds $\beta_2 = 2.54$: $p = \frac{e^{2.68}}{1+e^{2.68}} = 0.936$.

Answer

Sex: women had about 12.7 times the odds of survival of men ("women and children first"). **Class**: compared with 1st class, the odds were 73% lower in 2nd class (OR 0.27) and 92% lower in 3rd class (OR 0.075). **Age**: each extra decade reduced the odds by about 31% (OR 0.69). All effects are clearly non-zero.

Takeaway & common mistakes

Report odds ratios *and* a few predicted probabilities — probabilities are much easier to communicate (94% for a first-class woman vs. 8% for a third-class man).

Exercise 4.6 — A gamma regression for *T. rex* growth

Exercise 6 (exact text from the book)

The T. rex growth chart data plotted in Figure 3.10 has $n = 6$ observations with weights (kg) 29.9, 1761, 1807, 2984, 3230, 5040, and 5654 and corresponding ages (years) 2, 15, 14, 16, 18, 22 and 28. Since weight must be positive, the gamma family of distributions is a reasonable model for these data. Describe a model with gamma likelihood and log mean that increases linearly with age. Approximate the posterior using MCMC, summarize the posterior distribution of all model parameters, and plot the data versus the fitted mean curve.

In plain words

Model positive body masses with a gamma distribution whose (log) mean grows linearly with age, fit it by MCMC and draw the fitted growth curve.

Background you need

Gamma regression (mean–shape form). $Y_i \sim \text{Gamma}(\nu,\ \nu/\mu_i)$ (shape ν, rate ν/μ_i) has mean μ_i and variance μ_i^2/ν: the SD is proportional to the mean, natural for positive data. With $\log \mu_i = \beta_0 + \beta_1 \text{ age}_i$, e^{β_1} is the multiplicative change in mean mass per year.

Key idea

Three unknowns (β_0, β_1, ν); sample $(\beta_0, \beta_1, \log \nu)$ jointly by random-walk Metropolis (log scale keeps $\nu > 0$; remember the Jacobian $+ \log \nu$).

Solution

› Note on the data.

The exercise says $n = 6$ but lists 7 (mass, age) pairs (the same 7 as Listing 3.9); we use all 7.

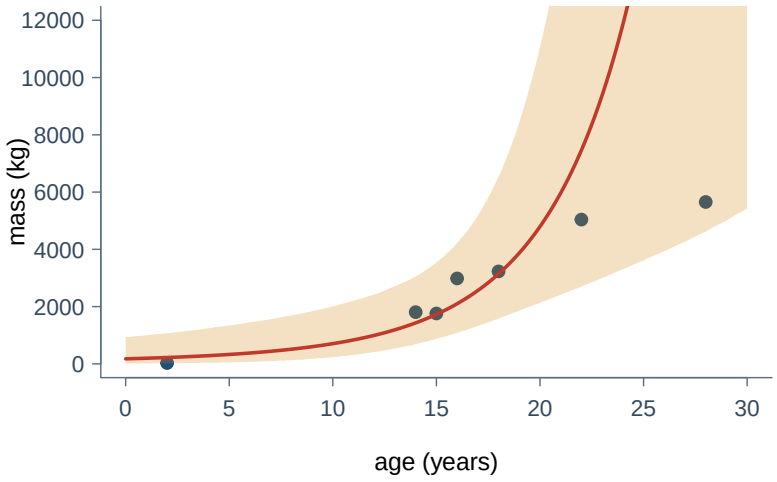
› Model.

$$\text{mass}_i \sim \text{Gamma}\big(\nu,\ \nu/\mu_i\big), \quad \log \mu_i = \beta_0 + \beta_1 \text{ age}_i, \quad \beta_0, \beta_1 \sim \text{N}(0, 10^2), \quad \nu \sim \text{Gamma}(0.1, 0.1).$$

› MCMC.

Three chains × 50,000 kept draws; acceptance 49%; $\hat{R} \leq 1.002$ for all parameters.

	Mean	SD	95% interval
β_0	4.40	1.11	(2.41, 6.84)
β_1	0.199	0.064	(0.066, 0.323)
ν (shape)	1.98	1.08	(0.50, 4.63)
growth per year $100(e^{\beta_1} - 1)$	22.2%	7.9%	(6.8%, 38.1%)



› Interpretation.

Mean mass grows by about 22% per year. A small shape ($\nu \approx 2$) means a large coefficient of variation ($1/\sqrt{\nu} \approx 0.7$), driven by the age-2 juvenile (29.9 kg) which the exponential curve over-predicts. The wide band shows how little 7 fossils can tell us.

Sanity check

The classical gamma GLM gives $\hat{\beta}_0 = 4.30$, $\hat{\beta}_1 = 0.199$: the same slope. ✓

Answer

Model: $\text{mass}_i \sim \text{Gamma}(\nu, \nu/\mu_i)$, $\log \mu_i = \beta_0 + \beta_1 \text{age}_i$. Posterior: $\beta_1 \approx 0.20$ (95% CI 0.07–0.32), i.e. $\approx 22\%$ growth per year; $\nu \approx 2$. The fitted curve follows the data but with wide uncertainty.

Takeaway & common mistakes

An exponential growth curve cannot level off; real dinosaur growth is sigmoidal. Compare the power-law curve in Exercise 3.17 — with 7 points, several shapes fit about equally well.

Exercise 4.7 — Gibbs sampler for the one-way random effects model

Exercise 7 (exact text from the book)

Consider the one-way random effects model $Y_{ij}|\alpha_i, \sigma^2 \sim \text{Normal}(\alpha_i, \sigma^2)$ and $\alpha_i \sim \text{Normal}(0, \tau^2)$ for $i = 1, \ldots, n$ and $j = 1, \ldots, m$. Assuming conjugate priors $\sigma^2, \tau^2 \sim \text{InvGamma}(a, b)$, derive the full conditional distributions of α_1, σ^2 and τ^2, and outline (but do not code) an MCMC algorithm to sample from the posterior.

In plain words

n groups, m observations each; each group has its own mean α_i, drawn from a common normal. Derive the three Gibbs updates and write the algorithm.

Background you need

Completing the square: a log density $-\frac{A}{2}x^2 + Bx$ means $x \sim \text{N}(B/A, 1/A)$. Inverse-gamma kernel: $x^{-a-1}e^{-b/x}$.

Key idea

Write the joint posterior once, then for each parameter keep only the factors that contain it.

Solution

Joint posterior.

$$p(\boldsymbol{\alpha}, \sigma^2, \tau^2 \mid \mathbf{Y}) \propto \prod_{i=1}^{n}\prod_{j=1}^{m}(\sigma^2)^{-\frac{1}{2}}e^{-\frac{(Y_{ij}-\alpha_i)^2}{2\sigma^2}} \prod_{i=1}^{n}(\tau^2)^{-\frac{1}{2}}e^{-\frac{\alpha_i^2}{2\tau^2}} (\sigma^2)^{-a-1}e^{-\frac{b}{\sigma^2}} (\tau^2)^{-a-1}e^{-\frac{b}{\tau^2}}.$$

Full conditional of α_1.

Factors with α_1: $-\frac{1}{2\sigma^2}\sum_j(Y_{1j}-\alpha_1)^2 - \frac{\alpha_1^2}{2\tau^2}$. Coefficient of α_1^2: $A = \frac{m}{\sigma^2} + \frac{1}{\tau^2}$; coefficient of α_1: $B = \frac{\sum_j Y_{1j}}{\sigma^2} = \frac{m\bar{Y}_1}{\sigma^2}$. So

$$\alpha_1 \mid \cdot \sim \text{N}\Big(\frac{m\bar{Y}_1/\sigma^2}{m/\sigma^2 + 1/\tau^2},\ \frac{1}{m/\sigma^2 + 1/\tau^2}\Big).$$

Why? The mean is a precision-weighted compromise between the group average $\bar{Y}_1$ and the prior mean 0: groups are shrunk toward 0 more when τ^2 is small or m is small.

Full conditional of σ^2.

Collect all nm likelihood terms and the prior:

$$p(\sigma^2 \mid \cdot) \propto (\sigma^2)^{-\frac{nm}{2}-a-1}\exp\Big\{-\frac{b + \frac{1}{2}\sum_{i,j}(Y_{ij}-\alpha_i)^2}{\sigma^2}\Big\},$$

so

$$\sigma^2 \mid \cdot \sim \text{InvGamma}\Big(a + \frac{nm}{2},\ b + \frac{1}{2}\sum_{i,j}(Y_{ij}-\alpha_i)^2\Big).$$

Full conditional of τ^2.

Only the n random-effect terms and its prior:

$$\tau^2 \mid \cdot \sim \text{InvGamma}\Big(a + \frac{n}{2},\ b + \frac{1}{2}\sum_{i=1}^{n}\alpha_i^2\Big).$$

Algorithm.

1. Initialise $\alpha_i^{(0)} = \bar{Y}_i$, $\sigma^{2(0)}$ = average within-group variance, $\tau^{2(0)}$ = variance of the $\bar{Y}_i$.
2. For $s = 1, \dots, S$: (i) for $i = 1, \dots, n$ draw $\alpha_i^{(s)}$ from its normal full conditional using $\sigma^{2(s-1)}, \tau^{2(s-1)}$; (ii) draw $\sigma^{2(s)}$ given $\boldsymbol{\alpha}^{(s)}$; (iii) draw $\tau^{2(s)}$ given $\boldsymbol{\alpha}^{(s)}$.
3. Discard burn-in; check trace plots, $\hat{R}$ (several chains) and ESS — particularly for τ^2, which mixes slowest when n is small.

Answer

$\alpha_1 \mid \cdot \sim \mathrm{N}\big(\frac{m\bar{Y}_1/\sigma^2}{m/\sigma^2+1/\tau^2}, \frac{1}{m/\sigma^2+1/\tau^2}\big)$; $\sigma^2 \mid \cdot \sim \mathrm{InvGamma}(a + \frac{nm}{2},\ b + \frac{1}{2}\sum_{ij}(Y_{ij} - \alpha_i)^2)$; $\tau^2 \mid \cdot \sim \mathrm{InvGamma}(a + \frac{n}{2},\ b + \frac{1}{2}\sum_i \alpha_i^2)$. Cycle through them (Gibbs).

Takeaway & common mistakes

This is exactly the sampler used (with covariates added) for the state effects in Exercise 3.

Exercise 4.8 — Malaria in Gambian children

Exercise 8 (exact text from the book)

Load the Gambia data in R:

```
> library(geoR)
> data(gambia)
> ?gambia
```

The response variable Y_i is the binary indicator that child i tested positive for malaria (pos), and the remaining seven variables are covariates.

(a) Fit the logistic regression model

$$\text{logit}[\text{Prob}(Y_i = 1)] = \sum_{j=1}^{p} X_{ij}\beta_j$$

with uninformative priors for the β_j. Verify that the MCMC sampler has converged, and summarize the effects of the covariates.

(b) In this dataset, the 2,035 children reside in $L = 65$ unique locations (defined by the x and y coordinates in the dataset). Let $s_i \in \{1, \ldots, L\}$ be the label of the location for observation i. Fit the random effects logistic regression model

$$\text{logit}[\text{Prob}(Y_i = 1)] = \sum_{j=1}^{p} X_{ij}\beta_j + \alpha_{s_i} \quad \text{where} \quad \alpha_l \overset{\text{iid}}{\sim} \text{Normal}(0, \tau^2)$$

and the β_j and τ^2 have uninformative priors. Verify that the MCMC sampler has converged; explain why random effects might be needed here; discuss and explain any differences in the posteriors of the regression coefficients that occur when random effects are added to the model; plot the posterior means of the α_l by their spatial locations, and suggest how this map might be useful to malaria researchers.

In plain words

Model which children test positive for malaria using age, bed-net use, insecticide treatment, greenness, a health-centre indicator and the village coordinates; then add village effects and map them.

Background you need

- Logistic regression and block Metropolis: primer, point 3.
- **Location-level covariates** (same value for every child in a village: `green`, `phc`, `x`, `y`) can only be learned from *differences between villages*.

Key idea

Children in the same village share mosquitoes, water and health care, so their outcomes are correlated. Ignoring this pretends we have 2,035 independent pieces of information about village-level covariates when we really have 65.

Solution

Covariates

(continuous ones standardised): age (years), `netuse` (sleeps under a bed net), `treated` (net treated with insecticide), `green` (satellite greenness), `phc` (village has a primary health centre), and the coordinates `x`,

y. Prevalence: 35.7%.

› (a) Logistic regression.

Three chains × 12,000 kept draws, acceptance 27%, all $\hat{R} \leq 1.002$.

	Mean	SD	95% interval	Odds ratio
intercept	0.161	0.114	$(-0.06, 0.38)$	—
age (per SD)	0.267	0.048	(0.17, 0.36)	1.31
netuse	-0.697	0.119	$(-0.93, -0.46)$	0.50
treated	-0.032	0.140	$(-0.32, 0.24)$	0.97
green (per SD)	-0.029	0.094	$(-0.21, 0.16)$	0.97
phc	-0.445	0.117	$(-0.67, -0.21)$	0.64
x (per SD)	0.386	0.092	(0.20, 0.56)	1.47
y (per SD)	-0.338	0.053	$(-0.44, -0.24)$	0.71

Older children are more likely to test positive; bed nets halve the odds; a health centre lowers the odds by a third; prevalence increases eastward and southward.

› (b) Why random effects?

Villages differ in many unmeasured ways (breeding sites, housing, distance to water). Children in one village therefore share a common risk: residuals are correlated within villages, and treating them as independent understates the uncertainty of village-level effects.

› (b) Model and sampler.

$\alpha_l \overset{\text{iid}}{\sim} \mathrm{N}(\mu, \tau^2)$ (centred), $\tau^2 \sim \mathrm{InvGamma}(0.1, 0.1)$. Each iteration: a block Metropolis step for $\boldsymbol{\beta}$, a Metropolis step for all 65 α_l at once (they are conditionally independent given $\boldsymbol{\beta}$), Gibbs draws for μ and τ^2. Three chains × 20,000 kept draws.

› (b) Convergence.

All $\hat{R} \leq 1.013$. ESS ranges from 230 (phc) to 1,700 (age), and 2,037 for τ. The village-level coefficients mix more slowly because they trade off against the village effects — precisely the confounding described above — but the ESS is adequate for posterior means and 95% intervals.

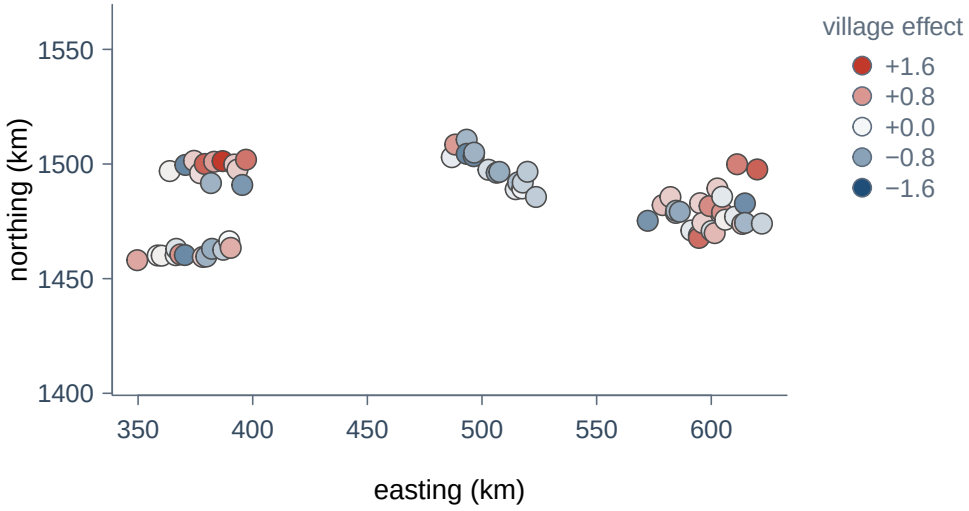
› (b) Effect on the coefficients.

	No random effects		Village random effects		
	mean	SD	mean	SD	95% interval
age	0.267	0.048	0.288	0.052	(0.19, 0.39)
netuse	-0.697	0.119	-0.505	0.160	$(-0.82, -0.19)$
treated	-0.032	0.140	-0.274	0.205	$(-0.67, 0.13)$
green	-0.029	0.094	-0.098	0.227	$(-0.55, 0.36)$
phc	-0.445	0.117	-0.458	0.232	$(-0.92, -0.00)$
x	0.386	0.092	0.503	0.229	(0.05, 0.95)
y	-0.338	0.053	-0.365	0.113	$(-0.58, -0.14)$
τ	—	—	0.795	0.106	(0.61, 1.03)

- **Child-level** covariates (age) barely change.
- **Village-level** covariates (green, phc, x, y): SDs roughly *double* — they are now (correctly) estimated from 65 villages rather than 2,035 children.
- The bed-net effect shrinks ($-0.70 \to -0.51$): part of the naive effect was villages with high net use also being low-risk villages.

- $\tau \approx 0.8$: substantial variation between villages (an odds ratio of $e^{2\times 0.8} \approx 5$ between a high-risk and a low-risk village two SDs apart).

› (b) Map of village effects.



Red villages have higher malaria risk than their covariates predict, blue ones lower (effects from -1.05 to $+1.58$ on the log-odds scale). Researchers can use this map to (i) target bed-net distribution and spraying at red "hotspots", (ii) look for environmental causes shared by neighbouring red villages (rice fields, river proximity), and (iii) design future surveys.

Answer

(a) Converged; age $+$, bed nets $-$ (OR 0.5), health centre $-$, east/south higher risk; insecticide treatment and greenness unclear. (b) Converged ($\hat{R} \leq 1.013$); random effects are needed because children within a village are correlated; adding them leaves child-level effects alone but roughly doubles the uncertainty of village-level effects and weakens the bed-net effect; $\tau \approx 0.8$. The map highlights high-risk villages for targeted intervention.

Takeaway & common mistakes

Whenever covariates are measured at a group level, a model without group effects is over-confident about them.

Exercise 4.9 — How popular will the name "Sophia" be?

Exercise 9 (exact text from the book)

Download the babynames data in R, and compute the log odds of a baby being named "Sophia" each year after 1950:

```
library(babynames)
dat ← babynames
dat ← dat[dat$name=="Sophia" &
          dat$sex=="F" &
          dat$year>1950,]
yr ← dat$year
p  ← dat$prop
t  ← dat$year - 1950
Y  ← log(p/(1-p))
```

Let Y_t denote the sample log-odds in year $t + 1950$. Fit the following time series (auto-regressive order 1) model to these data:

$$Y_t = \mu_t + \rho(Y_{t-1} - \mu_{t-1}) + \varepsilon_t$$

where $\mu_t = \alpha + \beta t$ and $\varepsilon_t \overset{\text{iid}}{\sim} \text{Normal}(0, \sigma^2)$. The priors are $\alpha, \beta \sim \text{Normal}(0, 100^2)$, $\rho \sim \text{Uniform}(-1, 1)$, and $\sigma^2 \sim \text{InvGamma}(0.1, 0.1)$.

(a) Give an interpretation of each of the four model parameters: α, β, ρ, and σ^2.

(b) Fit the model using JAGS for $t > 1$, verify convergence, and report the posterior mean and 95% interval for each parameter.

(c) Plot the posterior predictive distribution for Y_t in the year 2020.

In plain words

Model the year-by-year log-odds of the name Sophia as a linear trend plus persistent deviations, and forecast 2020.

Background you need

- **AR(1)**: $e_t = \rho e_{t-1} + \varepsilon_t$ with $|\rho| < 1$ is stationary — deviations decay back to 0. As $\rho \to 1$ it becomes a *random walk*, which never returns.
- Given ρ, the model is linear in (α, β): $Y_t - \rho Y_{t-1} = \alpha(1-\rho) + \beta\{t - \rho(t-1)\} + \varepsilon_t$. Given (α, β), it is a regression of $e_t = Y_t - \mu_t$ on e_{t-1} with slope ρ. So every full conditional is normal (truncated to $(-1, 1)$ for ρ) or inverse-gamma.

Key idea

μ_t is the long-run trend line; ρ says how long a deviation from the trend lasts. To forecast 2020 from the last observed year (2017), simulate three steps forward, once per posterior draw.

Solution

(a) Interpretation.

- α: the trend value of the log-odds at $t = 0$ (the year 1950).
- β: the trend slope — change in log-odds per year (e^β = yearly multiplicative change in the odds).
- ρ: persistence — the fraction of this year's deviation from the trend that carries over to next year.
- σ^2: variance of the year-to-year shocks (new, unpredictable changes in popularity).

› (b) Fit.

Data: 67 years (1951–2017), Sophia rising from a log-odds of about -9 to -4.8 (0.79% of girls in 2017). Our Gibbs sampler (three chains from $\rho = -0.5, 0, 0.9$; 25,000 kept draws each):

	Mean	SD	95% interval	$\hat{R}$	ESS
α	-10.3	65.9	$(-154,\ 143)$	1.000	72,307
β	0.083	0.148	$(-0.24,\ 0.41)$	1.000	72,023
ρ	0.987	0.023	$(0.912,\ 0.9998)$	1.000	7,681
σ	0.181	0.016	$(0.153,\ 0.216)$	1.000	70,927

Sanity check

JAGS output (run on GitHub Actions; 3 chains × 30,000 draws).

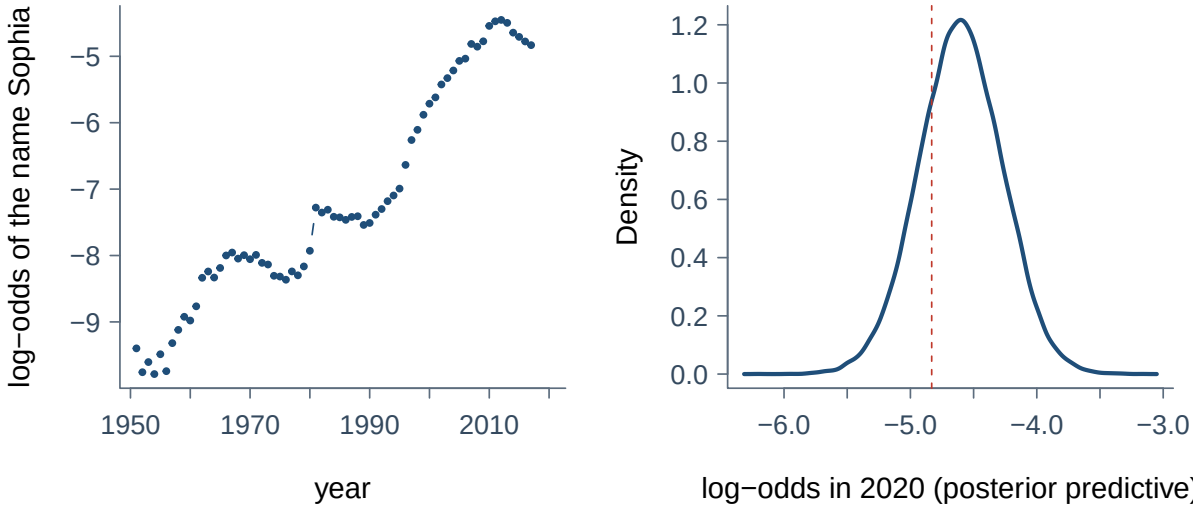
	Mean	95% interval	$\hat{R}$	ESS (of 90,000)
α	-10.9	$(-166,\ 140)$	1.006	920
β	0.082	$(-0.27,\ 0.44)$	1.011	580
ρ	0.987	$(0.913,\ 0.9998)$	1.006	599
σ	0.181	$(0.152,\ 0.216)$	1.000	77,870

Same posterior as our sampler. Note the much smaller ESS for α, β, ρ in JAGS: our Gibbs sampler updates (α, β) *jointly* given ρ, which removes their strong posterior correlation, whereas JAGS updates them one at a time.

Why? Why is α so uncertain? With $\rho \approx 1$, the trend enters the model only through $\alpha(1-\rho)$ and $\beta\{t - \rho(t-1)\}$, which vanish as $\rho \to 1$: the series behaves like a random walk and the trend line is almost unidentifiable. That is a genuine feature of these data, not a convergence failure ($\hat{R} = 1$, large ESS).

› (c) Posterior predictive for 2020.

For each posterior draw: start from Y_{2017} and apply the model three times, $Y_t = \mu_t + \rho(Y_{t-1} - \mu_{t-1}) + \varepsilon_t$ for $t = 68, 69, 70$.



PPD for 2020: log-odds mean -4.60, 95% interval $(-5.25,\ -3.95)$; on the proportion scale, median 0.99% of girls, 95% interval (0.52%, 1.88%).

Answer

(a) α = trend level in 1950, β = trend slope per year, ρ = persistence of deviations, σ^2 = shock variance. (b) $\rho \approx 0.99$ (95% CI 0.91–1.00), $\sigma \approx 0.18$; α and β are poorly determined because the series is close to a random walk; JAGS agrees. (c) 2020 log-odds ≈ -4.6 $(-5.25, -3.95)$, i.e. about 1% of girls (0.5–1.9%).

Takeaway & common mistakes

When $\rho \approx 1$, forecasts are driven by the most recent value, not by a long-run trend — and their uncertainty grows quickly with the horizon.

Exercise 4.10 — A B-spline curve for global temperature

Exercise 10 (exact text from the book)

The global mean temperature anomaly (difference between current year and the 1951-1980 average) data on the course website and plotted in Figure 3.4. Let X_i and Y_i be the year and temperature anomaly (C) for observation i. We will analyze these data using a nonparametric regression model $Y_i = g(X_i) + \varepsilon_i$ as in Section 8.3, where g is modeled as a linear combination of $J = 15$ B-spline basis functions.

(a) Fit the model, and plot the data along with the posterior mean and 95% posterior interval of $g(X_i)$ for each value X_i.

(b) Summarize and interpret the posterior distribution of $\Delta = g(2020) - g(1980)$.

(c) Summarize and interpret the posterior distribution for $\delta_i = g(X_i) - g(X_{i-1})$ for each $i > 1$.

In plain words

Fit a smooth curve to 142 yearly temperature anomalies, measure the warming between 1980 and 2020, and estimate the warming rate in each year.

Background you need

B-splines: J smooth "bump" functions $B_1(x), \ldots, B_J(x)$ spread over the range of x. Writing $g(x) = \sum_{j=1}^{J} \beta_j B_j(x)$ turns curve fitting into *linear regression* on the J columns $B_j(X_i)$ (R: `splines::bs(year, df = 15, intercept = TRUE)`). Any function of g (differences, maxima) has a posterior obtained by applying it to each draw of $\boldsymbol{\beta}$.

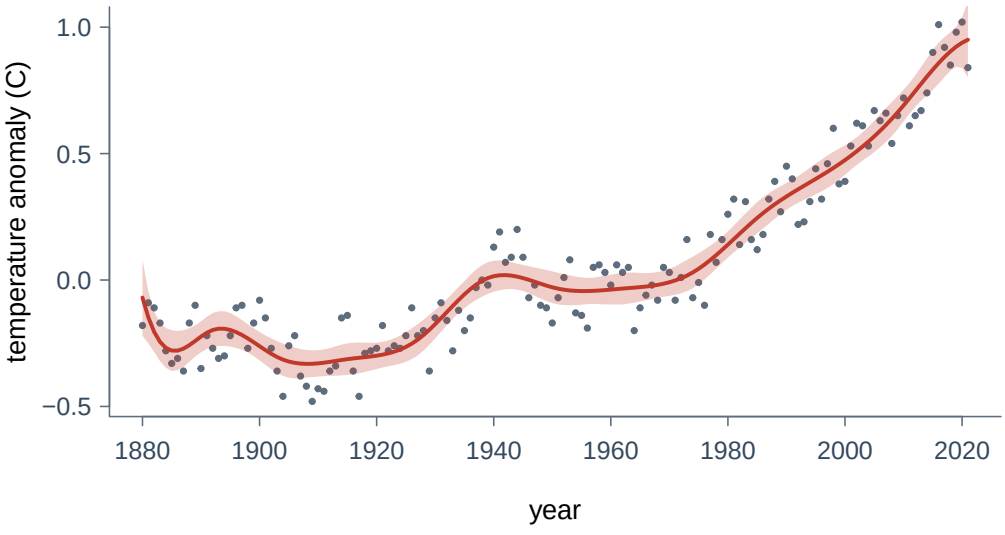
Key idea

Gibbs sampler for linear regression on the 15 basis columns; for each draw compute the whole curve $g(X_1), \ldots, g(X_{142})$, then Δ and the δ_i.

Solution

(a) Fit.

Data: 1880–2021. Priors $\beta_j \sim \mathrm{N}(0, 100^2)$, $\sigma^2 \sim \mathrm{InvGamma}(0.01, 0.01)$; 10,000 kept draws. Residual SD $\sigma \approx 0.099$°C.

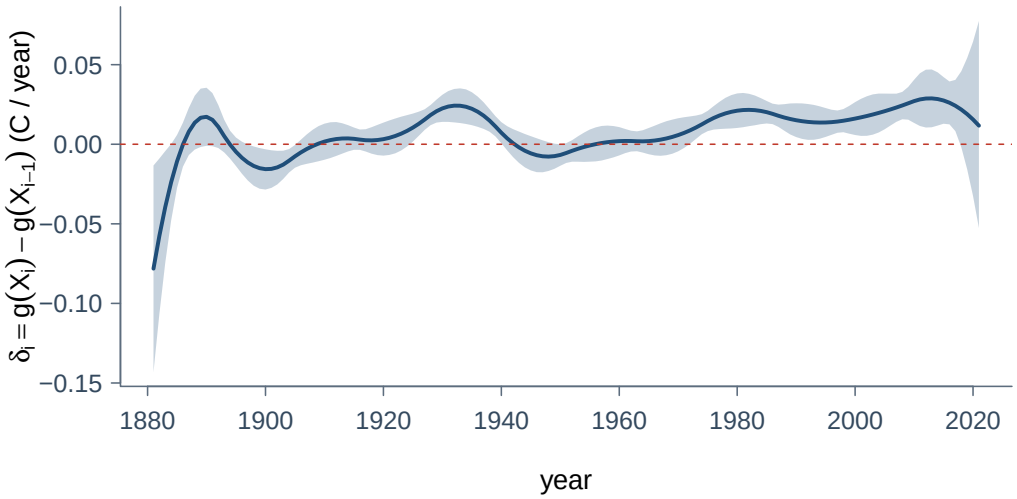


The curve shows slight cooling to $\approx$ 1910, warming to the 1940s, a plateau to the 1970s, then rapid warming.

› (b) $\Delta = g(2020) - g(1980)$.

For each draw, subtract the curve values at 1980 and 2020: posterior mean $\mathbf{0.80°C}$, 95% interval $(0.69,\ 0.91)$, $\text{Prob}(\Delta > 0 \mid \text{data}) > 0.9999$. The trend (not just the noisy yearly values) warmed by about 0.8°C in 40 years — roughly 0.2°C per decade.

› (c) Year-to-year changes δ_i.



The posterior mean rate peaks at 0.029°C/year (2013). In 67 of the 141 years the posterior probability of warming ($\delta_i > 0$) exceeds 0.95, including every year from 1972 to 2018. The interval for the last year (2021: 0.012, 95% CI $(-0.053,\ 0.077)$) is wide: splines are least certain at the ends of the data.

Answer

(a) See figure: a smooth curve with a narrow 95% band. (b) $\Delta \approx 0.80$°C (95% CI 0.69–0.91): warming between 1980 and 2020 is certain. (c) The warming rate has been positive with posterior probability above 0.95 in every year from 1972 to 2018, peaking around 0.03°C/year; rates for 2019–2021 are uncertain (edge effect).

Takeaway & common mistakes

Bayesian inference gives uncertainty for *any* derived quantity (differences, derivatives, maxima) for free — just compute it draw by draw.

Exercise 4.11 — When to use which prior: marine biodiversity

Exercise 11 (exact text from the book)

Consider an ecological study where the response, Y_i, is a measure of marine biodiversity at location $i = 1, \ldots, n$. The covariates $X_{i1}, \ldots, X_{ip}$ are summaries of the environment of location i, such as water depth, distance to shore, fishing regulations, etc. Assume the multiple linear regression model $Y_i = \beta_0 + X_{i1}\beta_1 + \cdots + X_{ip}\beta_p + \varepsilon_i$. Give a scenario where the following priors for the β_j might be useful:

(a) Jeffreys prior

(b) Bayesian LASSO prior

(c) An informative prior $\beta_j \sim \text{Normal}(m_j, v)$ for small v

Your scenarios should include both numerical summaries such as n and p and scientific objectives.

In plain words

Describe a realistic study in which each kind of prior is the natural choice.

Background you need

- **Jeffreys** (flat on $\boldsymbol{\beta}$, $1/\sigma^2$ on the variance): objective; reproduces least squares; needs $n \gg p$.
- **Bayesian LASSO** (double exponential): shrinks small coefficients toward 0; for many candidate predictors.
- **Informative** $\text{N}(m_j, v)$: encodes reliable outside knowledge.

Key idea

Match the prior to (i) how many covariates there are relative to the sample size and (ii) what is already known.

Solution

(a) Jeffreys prior.

Scenario: a large monitoring programme with $n = 800$ reef sites and $p = 4$ pre-specified covariates (depth, distance to shore, protection status, water temperature). *Objective*: an impartial estimate of the effect of marine protected areas, to be reported to a regulator. With $n/p = 200$ the data dominate and an objective prior avoids any accusation that the conclusions were "built into" the prior.

(b) Bayesian LASSO.

Scenario: $n = 60$ sites with $p = 45$ candidate environmental summaries (depth, slope, salinity, 10 nutrient measures, 20 satellite indices, fishing pressure, …), many of them correlated. *Objective*: identify which few factors drive biodiversity and build a predictive map. With p close to n least squares is unstable; the LASSO prior shrinks the many negligible effects toward zero and stabilises the important ones.

(c) Informative prior with small v.

Scenario: a small pilot study, $n = 12$ sites in a new region, $p = 3$ (depth, distance to shore, protection). A meta-analysis of 40 earlier studies estimated the depth effect precisely, giving m_{depth} and a small v. *Objective*: estimate the protection effect in the new region. With only 12 sites, borrowing the well-established depth effect frees the scarce data to inform the protection effect.

Answer

(a) Jeffreys: large n (e.g. 800), few pre-specified p (4), objective reporting. (b) Bayesian LASSO: p near n (45 vs 60), variable selection/prediction. (c) Informative: tiny new study ($n = 12$, $p = 3$) plus strong external evidence on some effects.

Takeaway & common mistakes

The prior is part of the study design: it should reflect the ratio p/n and the existing scientific knowledge.

Exercise 4.12 — Genetic markers for depression

Exercise 12 (exact text from the book)

In a study of the genetic determinants of depression, researchers gathered information from 1000 patients. The response variable is the (continuous) CES-D survey score and the predictors 10,000 genetic markers. The objective is to determine if any of the markers are associated with depression.

(a) Describe and justify a model and prior for these data

(b) How would you summarize the results?

In plain words

1,000 patients but 10,000 possible predictors: design a Bayesian model that can find the few markers (if any) that matter.

Background you need

Spike-and-slab prior: $\beta_j = \gamma_j\, b_j$ with $\gamma_j \sim \text{Bernoulli}(q)$ ("is marker j active?") and $b_j \sim \text{N}(0, v)$. The posterior inclusion probability $\text{Prob}(\gamma_j = 1 \mid \mathbf{Y})$ measures the evidence for marker j. Continuous alternatives: horseshoe or Bayesian LASSO priors.

Key idea

With $p = 10n$, least squares is impossible; the prior must encode *sparsity*: most markers have no effect.

Solution

(a) Model.

$$Y_i = \beta_0 + \sum_k W_{ik}\eta_k + \sum_{j=1}^{10{,}000} G_{ij}\beta_j + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \text{N}(0, \sigma^2),$$

where Y_i is the CES-D score (possibly log-transformed, since it is skewed and bounded), G_{ij} the standardised genotype (0/1/2 copies of the minor allele), and W_{ik} **confounders** that must not be shrunk: age, sex, and the first few genetic principal components (to adjust for ancestry, which otherwise creates spurious associations).

(a) Priors.

- Markers: spike-and-slab $\beta_j = \gamma_j b_j$, $\gamma_j \sim \text{Bernoulli}(q)$, $b_j \sim \text{N}(0, v)$, with $q \sim \text{Beta}(1, 1000)$ so that *a priori* only ≈ 10 markers are expected to be active, and the data can learn q (this automatically adjusts for multiplicity).
- Confounders and intercept: vague normal priors; σ^2: vague inverse-gamma.

Justification: genetic effects on complex traits are typically few and small; the sparsity prior prevents over-fitting and shares information across markers about how many are active.

(b) Summaries.

- A Manhattan-style plot of the posterior inclusion probabilities $\text{Prob}(\gamma_j = 1 \mid \mathbf{Y})$ along the genome.
- A list of markers with high inclusion probability, chosen to control the **Bayesian false discovery rate** (e.g. the largest set whose average $1 - \text{Prob}(\gamma_j = 1 \mid \mathbf{Y})$ is below 5%), with posterior means and 95% intervals of their effects.

- The posterior of q (how many markers are active) and of the variance explained by genetics.
- If nothing passes the threshold, say so: "no marker shows convincing evidence of association".

Answer

(a) Linear regression of (transformed) CES-D on standardised markers plus unshrunk confounders (age, sex, ancestry PCs), with a sparsity-inducing spike-and-slab (or horseshoe) prior and a learned inclusion rate. (b) Posterior inclusion probabilities (plot along the genome), a Bayesian-FDR-controlled list of markers with effect estimates, and the posterior number of active markers.

Takeaway & common mistakes

In high dimensions, the prior *is* the regularisation. Learning the sparsity level from the data is what makes the Bayesian approach handle multiple testing automatically (Chapter 5).

Exercise 4.13 — A three-component normal mixture for the galaxies

Exercise 13 (exact text from the book)

Open and plot the galaxies data in R using the code below,

```
> library(MASS)
> data(galaxies)
> ?galaxies
> Y ← galaxies
> hist(Y,breaks=25)
```

Model the observations $Y_1, \ldots, Y_{82}$ using a mixture of $K = 3$ normal distributions. For each of the S MCMC iterations evaluate the density function on the grid $y \in \{5000, 5100, \ldots, 40000\}$ (351 points in the grid), giving an $S \times 351$ matrix of posterior samples. Plot the posterior median and 95% credible set of the density function at each of the 351 grid values. Does this mixture model fit the data well?

In plain words

Fit a mixture of three bell curves to the galaxy velocities and show the fitted density with its uncertainty.

Background you need

Mixture model with latent labels: $z_i \in \{1, 2, 3\}$ says which component galaxy i belongs to; $\text{Prob}(z_i = k) = w_k$, $Y_i \mid z_i = k \sim \text{N}(\mu_k, \sigma_k^2)$. Gibbs sampler: (1) $\text{Prob}(z_i = k \mid \cdot) \propto w_k \phi(Y_i; \mu_k, \sigma_k^2)$; (2) $\mathbf{w} \mid \mathbf{z} \sim \text{Dirichlet}(1 + n_1, 1 + n_2, 1 + n_3)$; (3) μ_k, σ_k^2 updated from the galaxies currently in component k (normal and inverse-gamma).

Key idea

The density $f(y) = \sum_k w_k \phi(y; \mu_k, \sigma_k^2)$ is a function of the parameters, so each MCMC draw gives one whole curve. Percentiles across the S curves give the band. (Label switching does not matter for the density.)

Solution

Set-up.

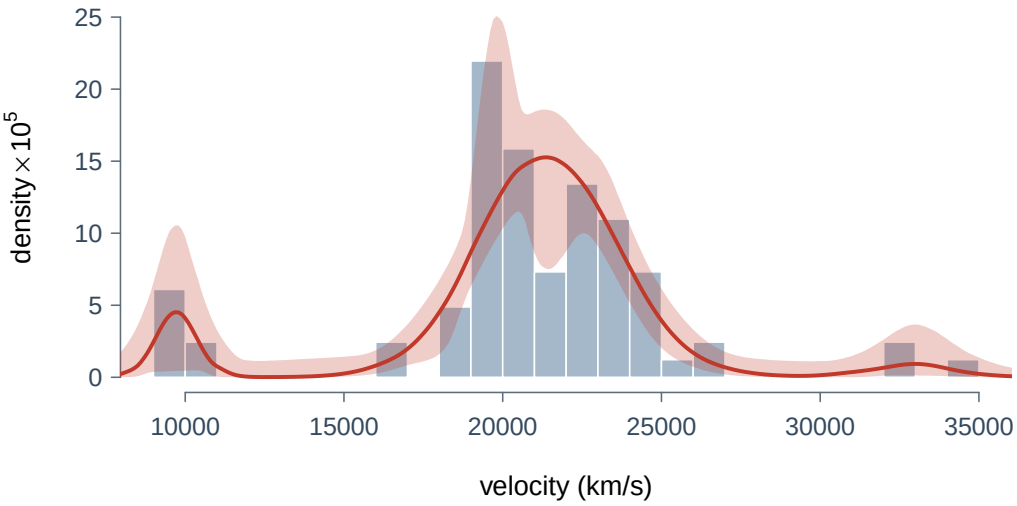
Velocities in 1,000 km/s. Priors: $\mu_k \sim \text{N}(20, 10^2)$, $\sigma_k^2 \sim \text{InvGamma}(1, 1)$, $\mathbf{w} \sim \text{Dirichlet}(1, 1, 1)$. 20,000 iterations, 5,000 burn-in: an $S \times 351$ matrix with $S = 15{,}000$.

Component estimates

(sorted by mean):

Component	mean μ_k (1000 km/s)	SD σ_k	weight w_k
1	9.72 (9.17, 10.28)	0.71	0.09
2	21.37 (20.84, 21.90)	2.20	0.85
3	32.29 (25.30, 34.48)	1.77	0.06

Fitted density.



› Does it fit?

Much better than the single t (Exercise 3.9) or Laplace (Exercise 3.10): the small clusters near 10,000 and 33,000 km/s now get their own components. It is still not perfect: the large central component smooths over visible structure between 16,000 and 26,000 km/s (the histogram has two or three bumps there), so $K = 4$–6 would likely fit better.

Answer

The three components sit at about 9,700, 21,400 and 32,300 km/s with weights 0.09, 0.85, 0.06. The posterior median density and 95% band track the histogram well, including both outer clusters; the remaining misfit is sub-structure inside the main cluster, suggesting more components.

Takeaway & common mistakes

Mixtures are a flexible way to model multi-modal data. Choosing K is a model-selection question (Chapter 6).

Exercise 4.14 — Designing a habitat model for woodpeckers

Exercise 14 (exact text from the book)

A group of 10 ecologists is surveying a forest for Red Cockaded Woodpeckers (RCP). Each ecologist will walk along a different path and make 25 stops. At each stop, they will record local conditions (tree density, elevation, etc.) and whether they see or hear an RCP. The objective is to build a model for the types of habitat that are the most favorable to the RCP.

(a) Describe and justify a model and prior for these data

(b) How would you summarize the results?

In plain words

250 yes/no observations (10 paths × 25 stops) with habitat covariates. Design a Bayesian model that finds the best habitat while accounting for who did the survey and where.

Background you need

- Binary outcomes ⇒ logistic regression.
- Stops on the same path (same ecologist, nearby locations) are correlated ⇒ random effects (Exercise 8).
- **Imperfect detection** (Exercise 1.15): "not seen" does not mean "absent".

Key idea

Separate *where the birds live* (habitat) from *how good each ecologist is at spotting them* (detection).

Solution

(a) Model.

For stop j on path i, $Y_{ij} = 1$ if an RCP is seen or heard. A simple and defensible choice is a logistic mixed model

$$\operatorname{logit} \operatorname{Prob}(Y_{ij} = 1) = \beta_0 + \sum_k X_{ijk}\beta_k + u_i, \qquad u_i \overset{\text{iid}}{\sim} \mathrm{N}(0, \tau^2),$$

with X_{ijk} the standardised habitat covariates (tree density, elevation, tree age, understory, …). The path/ecologist effect u_i absorbs differences in observer skill and in the part of the forest each path covers. Because consecutive stops are close in space, one can add a spatially correlated term (e.g. an AR(1) effect along the path) if residuals are correlated.

Better (occupancy) version: write $Y_{ij} = Z_{ij}D_{ij}$ with occupancy $Z_{ij} \sim \text{Bernoulli}(\psi_{ij})$, $\operatorname{logit}\psi_{ij} = \mathbf{X}_{ij}\boldsymbol{\beta}$ (habitat), and detection $D_{ij} \sim \text{Bernoulli}(p_i)$ with an ecologist-specific detection probability p_i. Repeat visits to some stops make ψ and p separately identifiable.

(a) Priors.

$\beta_k \sim \mathrm{N}(0, 2.5^2)$ on standardised covariates (weakly informative: odds ratios beyond e^5 per SD are implausible, which also prevents separation problems with 250 binary outcomes); $\tau \sim$ half-normal(0,1) or $\tau^2 \sim \text{InvGamma}(0.1, 0.1)$; detection $p_i \sim \text{Beta}(2, 2)$. If a previous RCP study exists, its estimates can inform the habitat priors.

(b) Summaries.

- Posterior means and 95% intervals of the odds ratios e^{β_k} for each habitat variable, and $\text{Prob}(\beta_k > 0 \mid \mathbf{Y})$.

- The posterior of τ (how much paths/observers differ) and of each ecologist's detection probability.
- Predicted occupancy probability ψ as a function of the key habitat variables (partial-effect curves), and a map of predicted suitability over the forest, with uncertainty.
- Posterior predictive checks (e.g. number of detections per path) to confirm the model fits.

Answer

(a) A logistic regression of detection on standardised habitat covariates with a random effect for path/ecologist (ideally an occupancy model separating habitat-driven occupancy from observer-specific detection), with weakly informative $\mathrm{N}(0, 2.5^2)$ priors. (b) Odds ratios with intervals, observer/path variation, predicted occupancy curves and a habitat-suitability map with uncertainty, plus posterior predictive checks.

Takeaway & common mistakes

Survey data carry the fingerprints of the people who collected them; a good model separates the biology from the observation process.